MASON CAO

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EDUCATION

Lambert High School, Suwanee, GA Expected May 2027

- Weighted GPA 4.615 | 27/866 | SAT 1530 | 16 AP and 9 dual-enrollment courses
- Georgia Tech dual enrollment: CS 1301 Intro to Computing, CS 1331 Object-Oriented Programming, Linear Algebra, Multivariable Calculus. Georgia State University dual enrollment: Introduction to Information Systems, Quantitative Reasoning, and other electives. University of North Georgia: CYBR 2050 Cybersecurity.

RESEARCH

Lead Researcher, Project AERIS 2025 to Present

Mentored by Dr. Annalisa Bracco, Georgia Tech Earth and Atmospheric Sciences; Senior Scientist, CMCC; Lester Mackey, Senior Principal Researcher at Microsoft Research and Adjunct Professor at Stanford University; and Dr. Greg Huey, Chair of the School of Earth and Atmospheric Sciences at Georgia Tech

- Built a verification layer testing whether physics-aware cross-source corroboration can evaluate AI air-quality claims without expert labeling, checking model explanations against ground, weather, and satellite data.
- First-author paper accepted to the NeurIPS 2026 main track; accepted for presentation at the AGU 2026 Annual Meeting; initiated the mentorship by cold-emailing 20 faculty.

Research Intern, George Mason University Exoplanet Research Summer 2026

- Applied Bayesian, time-series, and other statistical methods to NASA TESS and JWST data under Dr. Peter Plavchan; first-authored manuscript on refining characterization methods for TOI-3505.01. First author of seven: (DOI 10.13021/84125).

Selected Scholar, NLP Track, Stanford Pre-Collegiate Summer Institutes (SAIL) Summer 2026

- Built GPT-style language models in Hugging Face; solo capstone browser-focus classifier using Word2Vec and logistic regression with leakage-safe day-grouped splits.

LEADERSHIP AND SERVICE

Secretary and Director of Operations, First Step (student-founded environmental nonprofit) 2023 to Present

- Co-led a $5,000 Bloomberg grant funding Georgia Tech microplastics research; scaled 300+ volunteers across 250+ events.

Founder and Curriculum Director, Sharon Forks Library Summer Coding Camp Summers 2023 and 2024

- Founded a free CS camp and wrote its curriculum; taught 52 students Scratch, logic, and debugging (104 projects shipped).

Four-Year Varsity Letter Recipient, Lambert HS Swim and Dive; Senior Select, Swim Atlanta 2023 to Present

- 3-time GHSA 6A state qualifier; USA Swimming AA cuts; raced Speedo Best of the South and Georgia Senior State.

Section Leader and Principal Bass Clarinet, Lambert HS Wind Ensemble 2023 to Present

- Lead sectionals and mock All-State auditions for younger clarinetists; invited to perform at the Midwest Clinic and National Concert Band Fest.

WORK EXPERIENCE

Pool Manager, AMS Pools, Fulton and Forsyth County, GA Summers 2023 to Present

- Promoted from lifeguard to head guard to 1 of 6 managers over 8 pools and 100+ guards; scheduling, chemical logs, CPR drills. $20.50/hr.

SELECTED TECHNICAL PROJECTS

Green Spark AI: turns school energy, water, waste, and food data into a ranked, costed climate-action plan. USAII 2026.
FreshTrack: deployed pantry app cutting food waste with barcode scanning, expiry analytics, and recipes. 100+ users.
Nova Core: framework orchestrating 8 specialized AI agents for structured data and corporate workflows.
Detox: macOS focus tool shipped via Homebrew and DMG; island-survival mechanic tied to screen-time goals.
Kaggle notebooks: published machine-learning analyses of environmental sensor data. Forked 56+ times.

HONORS AND AWARDS

- **The Earthshot Prize 2026 (The Royal Foundation), Clean Our Air:** advanced to the longlist (Top 150 of 2,500+ global nominations).
- **Responsible AI Award,** one of five high school winners from 808 qualifying teams, USAII Global AI Hackathon 2026.
- **USACO Gold Division (2x),** Grades 10 and 11. Highest gold score: 715/1000. Silver promotion score: 890/1000.
- **Codeforces Candidate Master,** peak rating 1908.
- **7th Place, Networking (IT) Infrastructures,** FBLA National Leadership Conference.
- **USESO High Honors,** US Earth Science Olympiad National Open Exam.
- **DECA State Champion,** Entrepreneurship Team Decision Making; ICDC International Qualifier.
- **5x GMEA All-State Musician,** Soprano and Bass Clarinet.
- **Gold Medal, International Environmental Science Olympiad,** Top 10% Globally.
- **College Board Awards,** 2x AP Scholar with Distinction, 2x National Recognition Award.
- **President's Volunteer Service Award (PVSA)**, 2x Gold Medalist.

AFFILIATIONS AND TECHNICAL SKILLS

Affiliations: Google Developer Program; National Technical Honor Society; Spanish Honor Society (peer tutor); Key Club
Languages: Python, C++, Java, C, TypeScript, JavaScript, SQL.